MATH 350-2 Advanced Calculus

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Outline

1. Real Analysis Lecture 21
   - Infinite series
   - Infinite products

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Basic definition

Definition

An **infinite series** is an ordererd pair of sequences $(\{a_n\}, \{s_n\})$ where

$$s_n = a_1 + a_2 + \cdots + a_n.$$

The sequence $\{s_n\}$ is called the **sequence of partial sums** and s_n is called the n'th partial sum. We say the series converges or diverges if $\{s_n\}$ converges or diverges.

Notation:

$$a_1 + a_2 + a_3 + \ldots \quad \text{or} \quad \sum_{n=1}^{\infty} a_n.$$

Basic definition

We write

$$\sum_{n=1}^{\infty} a_n = L$$

to mean that the limit of the sequence s_n is L. We call L the **sum** of the series.

Examples

$$\sum_{n=1}^{\infty} \frac{1}{n^2+n} = 1.$$

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Telescoping:

$$\begin{aligned} s_n &= \frac{1}{1^2+1} + \frac{1}{2^2+2} + \frac{1}{3^2+3} + \cdots + \frac{1}{n^2+n} \\ &= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1}. \end{aligned}$$

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$$s_n = r + r^2 + r^3 + \cdots + r^n = r(1 + r + r^2 + \cdots + r^{n-1}) = r(1 + s_{n-1})$$

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$$s_n = \frac{r - r^{n+1}}{1 - r}$$

Examples

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n} = \ln(2)$$

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$$\ln(1-x) = -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{1}{4}x^4 - \dots$$

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Converges for $-1 \leq x < 1$.

Tests for Convergence

Theorem (Divergence Test)

If $\lim_{n\to\infty} a_n$ *either does not exist or is equal to something nonzero, then the series* $\sum a_n$ *diverges.*

Tests for Convergence

Theorem (Cauchy condition)

The series $\sum a_n$ converges if and only if for all $\epsilon > 0$ there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies

$$|a_n + a_{n+1} + \cdots + a_{n+p}| < \epsilon$$

for all $p \in \mathbb{Z}_+$.

Tests for Convergence

Definition

An **alternating series** is a series of the form $\sum(-1)^{n+1}a_n$ with $a_n \geq 0$ for all n.

Theorem (Alternating Series Test)

If a_n is a decreasing sequence converging to 0, then $\sum(-1)^{n+1}a_n$ converges to a real number s and

$$|s - s_n| < a_{n+1} \quad \forall\, n \in \mathbb{Z}_+.$$

Tests for Convergence

Definition

We say $\sum a_n$ is **absolutely convergent** if $\sum |a_n|$ converges.
We say $\sum a_n$ is **conditionally convergent** if it is convergent but not absolutely convergent.

Theorem (Absolute Convergence Test)

If a series is absolutely convergent, then it is convergent.